

Project 02: Multipartite Nonlocal Game

What is the highest success probability of the CHSH game if Alice and Bob share an entangled state? Provide an optimal protocol.

1 Alice-Bob Setup & Maximally Entangled State

Alice and Bob: Two distant observers. Each chooses one of two measurements (A_0, A_1 for Alice; B_0, B_1 for Bob). Each measurement gives outcomes ± 1 .

Alice

Measurements: A_0, A_1

Outcomes: ± 1

Bob

Measurements: B_0, B_1

Outcomes: ± 1

Maximally Entangled State:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|00\rangle + |11\rangle)$$

This gives maximum quantum correlations, enabling higher success probability than classical physics.

2 Success Probability

Success probability depends on correlators $\langle A_i B_j \rangle$. Higher correlations $\rightarrow$ higher success.

3 Correlators (Core Concept)

A correlator measures how strongly Alice's and Bob's outcomes are related:

$$\langle A_i B_j \rangle = \langle \psi | A_i \otimes B_j | \psi \rangle$$

Where: A_i = Alice's observable, B_j = Bob's observable, $|\psi\rangle$ = shared state, $\otimes$ = tensor product

Meaning: +1 (perfectly correlated), 0 (uncorrelated), -1 (perfectly anti-correlated)

4 G-Matrix (Correlation Matrix)

The G matrix collects all correlators:

$$G = \begin{bmatrix} \langle A_0 B_0 \rangle & \langle A_0 B_1 \rangle \\ \langle A_1 B_0 \rangle & \langle A_1 B_1 \rangle \end{bmatrix}$$

Why important: Contains all observable correlations, allows geometric interpretation, used to optimize success probability.

5 Representing Correlators as Vectors

From G matrix:

- **Alice's vectors:** u_0, u_1 (rows of G)
- **Bob's vectors:** v_0, v_1 (columns of G)
- **Unit circle constraint:** $||u_i|| = ||v_j|| = 1$

6 Unit Circle & Angles

Any unit vector in 2D:

$$(\cos \theta, \sin \theta)$$

Each correlator vector corresponds to an angle θ . The angle represents measurement direction; relative angle between Alice & Bob determines correlation strength.

7 Observables (Quantum Measurements)

Observables are operators. For qubits, we use Pauli matrices:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

8 Observable from Angle (Key Formula)

Once angle θ is known, the observable is:

$$A(\theta) = \cos \theta \cdot X + \sin \theta \cdot Z$$

Why: X and Z form a basis; angle θ fixes the measurement direction on the Bloch sphere.

9 Computing Correlators from Observables

Plug observables into:

$$\langle A_i B_j \rangle = \langle \psi | A_i \otimes B_j | \psi \rangle$$

This gives theoretical correlator values that should match the G matrix.

10 Final Logic Chain

Complete Flow:

1. Alice & Bob share maximally entangled state
2. Compute G matrix
3. G matrix $\rightarrow$ unit vectors (correlators)
4. Vectors $\rightarrow$ angles on unit circle
5. Angles $\rightarrow$ observables
6. Observables $\rightarrow$ correlators
7. Correlators $\rightarrow$ success probability

✨ Key Takeaway

Quantum correlations can be treated geometrically as unit vectors, whose angles define measurement observables that maximize Alice–Bob success probability when sharing a maximally entangled state.

Summary of Key Formulas

Entangled State:

$$|\psi\rangle = 1/\sqrt{2} (|00\rangle + |11\rangle)$$

Correlator:

$$\langle A_i B_j \rangle = \langle \psi | A_i \otimes B_j | \psi \rangle$$

G-Matrix:

$$G = \begin{bmatrix} \langle A_0 B_0 \rangle & \langle A_0 B_1 \rangle \\ \langle A_1 B_0 \rangle & \langle A_1 B_1 \rangle \end{bmatrix}$$

Observable from Angle:

$$A(\theta) = \cos \theta \cdot X + \sin \theta \cdot Z$$

Unit Vector:

$$(\cos \theta, \sin \theta)$$

CHSH Value:

$$S = |\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle|$$

Tsirelson Bound:

$$S_{\text{quantum}} \leq 2\sqrt{2} \approx 2.828$$

Classical Limit:

$$S_{\text{classical}} \leq 2$$

Calculating Success Probability for Optimal Protocol

Step 1: Calculate Optimal Value (V_q)

First, compute the CHSH value from correlators:

$$V_q = |\langle A_0 B_0 \rangle + \langle A_0 B_1 \rangle + \langle A_1 B_0 \rangle - \langle A_1 B_1 \rangle|$$

For optimal quantum strategy with maximally entangled state:

$$V_q = 2\sqrt{2} \approx 2.828$$

Step 2: Calculate Success Probability

The success probability formula is:

$$P_{\text{success}} = 1/2 (1 + V_q/2^n)$$

Where n is the number of measurement settings per party ($n = 2$ for CHSH).

Example Calculation:

For optimal quantum strategy where $V_q = 2\sqrt{2}$:

$$\begin{aligned} P_{\text{success}} &= 1/2 (1 + (2\sqrt{2})/2^2) \\ &= 1/2 (1 + (2\sqrt{2})/4) \\ &= 1/2 (1 + \sqrt{2}/2) \\ &\approx 1/2 (1 + 0.707) \\ &\approx 0.854 \text{ or } 85.4\% \end{aligned}$$

Compare with classical limit where $V_{\text{classical}} = 2$:

$$P_{\text{classical}} = 1/2 (1 + 2/4) = 1/2 (1 + 0.5) = 0.75 \text{ or } 75\%$$

Answer to Project Question:

The **highest success probability** of the CHSH game when Alice and Bob share a maximally entangled state is approximately **85.4%**.

The **optimal protocol** uses:

- Maximally entangled state: $|\psi\rangle = 1/\sqrt{2} (|00\rangle + |11\rangle)$
- Optimal measurement angles achieving $V_q = 2\sqrt{2}$
- Success probability: $P \approx 85.4\%$ (vs. classical limit of 75%)

Connection to Code: Your code automates this process by computing G matrix numerically, extracting unit vectors, finding optimal angles, constructing observables, verifying correlations, calculating V_q , and then computing success probability.

Quantum Information & Bell-Inequality Physics | CHSH-Type Correlations